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Mnemonic State Graph (MSG): A Cue-Gated, Human-in-the-Loop Memory Architecture for Conversational AI Systems

A Mechanism-Level Ablation Across 132 Randomized World Instances

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Note: This paper reports a mechanism-level synthetic ablation, not a validated human-subjects study. Results should be interpreted as validating structural claims about the MSG architecture rather than the system as a whole.

Keywords: long-term memory architecture, human-in-the-loop AI, cued recall, retrieval-augmented generation, conversational AI, cognitive psychology, confidence decay, dependency graph

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Abstract

Large language model (LLM) systems that maintain state across long interactions currently rely on one of three strategies: expanding the context window (linear cost growth, no forgetting), retrieval-augmented generation (RAG; stateless similarity search with no update semantics), or ad hoc paging (deciding what to keep in-context without a principled signal for what matters). We introduce the **Mnemonic State Graph (MSG)**, an architecture built on the thesis that cognition in a human–LLM system is *distributed*: the model's role is not to remember everything, but to manage a small set of pointers and to trigger human recall at the moments it is useful.

MSG combines four core innovations: (1) a *dual-graph working memory* — a strict dependency DAG for invalidation and a cyclic association graph for cueing and priming; (2) a *confidence model* that decays only on ignored cues rather than elapsed time; (3) a *six-type cue taxonomy* with differentiated human-acceptance dynamics; and (4) a *Socratic conflict-resolution protocol* that routes contradictory updates through explicit human confirmation rather than silent overwrite.

In a controlled synthetic evaluation across 11 independent runs (132 randomized world instances), MSG achieved higher recall accuracy than naive RAG (mean +22.8 pp) while using $4.8\times$ fewer tokens per turn. A companion LLM-in-the-loop pilot found correct discrimination on all 6 cue judgments. **These results validate the mechanism, not the system** — we report them transparently as a synthetic, mechanism-level ablation and enumerate explicit limitations.

Keywords: long-term memory architecture, human-in-the-loop AI, cued recall, retrieval-augmented generation, conversational AI, cognitive psychology, confidence decay, dependency graph

1. Introduction

1.1 Problem Statement

Modern LLM deployments face a fundamental tension: context windows are expensive and lossy at scale, but the alternatives are not obviously better. A 128k-token context window forces the model to re-process the entire history on every turn, with cost growing linearly and relevant information increasingly diluted among irrelevant tokens. The “lost in the middle” effect is now well documented^[1]. Retrieval-augmented generation (RAG)^[2] avoids the linear cost but introduces a different failure mode: it treats every stored chunk as equally valid regardless of whether it has since been superseded, contradicted, or corrected. Paging-based systems such as MemGPT/Letta^[3] improve on raw context expansion but still lack a principled signal for what to keep resident and what to evict.

We take as a starting premise that none of these architectures use the one resource available in essentially every real deployment: the human on the other end of the conversation, who already remembers most of what was discussed and only needs the right cue to retrieve it. Most memory architectures treat the human purely as a source of queries. We treat the human as a second, much larger and much cheaper memory store, and the LLM's job as being an *indexer and activator* of that store, not a competitor to it.

1.2 Thesis

Cognition in a deployed LLM system is distributed across three components: the model, the human, and whatever persistent storage backs the session. The correct objective for a memory subsystem is therefore not “minimize tokens” or “maximize retrieval recall” in isolation, but to *reduce human cognitive load while preserving task continuity*, using the model's limited working memory to decide when to activate the human's much larger one.

1.3 Contributions

1. **The MSG architecture:** A dual-graph working-memory model with a confidence model driven by structural importance, missed-cue decay, and recency; a six-type cue taxonomy; progressive hydration; and Socratic conflict resolution.
2. **A controlled synthetic benchmark:** Isolating three structural mechanisms against full-context and naive-RAG baselines, with fully randomized world generation (132 instances across 11 runs).
3. **A replication study:** Confirming the core effect is stable across randomized conditions, not a seed artifact.
4. **An LLM-in-the-loop pilot:** Testing cue discrimination with a real language model as judge, including negative controls.
5. **Explicit limitations:** Distinguishing what has and has not been established.

2. Related Work

Retrieval-Augmented Generation. RAG^[2] established embedding-based retrieval over an external store. Its retrieval trigger fires on every turn regardless of whether the human already knows the answer, and it has no native concept of a chunk being superseded. Long-context transformers^[1] have pushed raw context to 128k–1M tokens, but the “lost in the middle” effect persists and cost grows linearly.

Paging Systems. MemGPT/Letta^[3] introduced OS-inspired virtual context management. MSG's Layer 2 is architecturally similar but differs in trigger: MemGPT's paging decisions are model-internal, whereas MSG's cue engine externalizes the decision to the human via a dialogue act.

Generative Agent Memory. Generative Agents^[4] proposed a memory stream with importance scoring, but without dependency structure, human-confirmation loops, or explicit contradiction resolution.

Production Systems. Mem0^[5] and Zep^[6] are recent production memory layers. Zep uses a temporal knowledge graph; Mnemosyne^[7] uses graph-structured storage with temporal decay. The most methodologically mature comparison is Kerestecioglu^[8], with six biologically-grounded mechanisms evaluated on real benchmarks.

Cued Recall. MemoCue^[9] fine-tunes LLMs to transform queries into cue-rich forms for human recall. This is substantively different from MSG: MemoCue's cues help retrieve information the agent *lacks*; MSG's cues confirm information the agent *has*, using human response as an integrity check.

Positioning. MSG's distinguishing claims are: (a) treating the human as an integrity check via Socratic GC; (b) strict separation between dependency DAG and association graph; and (c) audit/tombstone design defending against gaslighting. None are validated against standard benchmarks yet.

3. The MSG Architecture

3.1 System Overview

MSG is organized into four layers. Information flows upward through the stack on recall and downward on creation. By design, the large majority of conversational turns never touch Layer 3 (the vector store) at all.

Table 1: MSG's Four-Layer Architecture

Layer	Name	Function
Layer 1	LLM + Cue Engine	Natural language interface, cue generation, acceptance handling
Layer 2	VPC (Variable Persistence Cache)	Dual-graph working memory (DAG + association)
Layer 3	Vector Store	Immutable, content-addressed embeddings
Layer 4	Session Store	On-disk serialization and audit logging

Key Insight

The key architectural bet of MSG is that most conversational turns do not require retrieval from a vector store. The working memory (Layer 2) maintains human-readable summaries and cues that resolve the majority of queries without ever touching Layer 3.

3.2 The VPC Node

Every salient concept becomes a **VPC node** with both human-facing (cue, summary) and machine-facing (embedding, graph topology, cognitive state) fields. The cue and summary are authored at creation time and fixed thereafter (only versioned on update), ensuring that the cue presented to the human is exactly the cue that was encoded, satisfying Tulving's encoding-specificity principle.

3.3 Dual Graph Topology

The Dependency DAG

The dependency DAG is strict and acyclic. It drives three mechanisms: (1) **Invalidation cascades**: updating a node discounts its children's confidence by $\gamma = 0.9$:

$$C_{\text{base}}^{(B)} \leftarrow C_{\text{base}}^{(B)} \times \gamma \quad (1)$$

(2) **Centrality computation**: PageRank^[21] assigns higher centrality to hub-like facts. (3) **Safe deprecation**: nodes with more than two dependents cannot be silently deprecated.

The Association Graph

The association graph is undirected, cyclic, and weighted by co-occurrence rate and embedding similarity. It drives cueing and priming. This draws on *spreading-activation theory*^{[10][11]}: activating one concept raises the accessibility of related concepts via associative links^[12].

3.4 Confidence Model

Confidence combines three multiplicative terms:

$$C_t = C_{\text{base}} \times C_{\text{structural}} \times C_{\text{recency}} \quad (2)$$

Base decay: C_{base} decays only on ignored cues:

$$C_{\text{base}} = C_0 \cdot \lambda^{N_{\text{missed}}}, \quad \lambda = 0.95 \quad (3)$$

This departs from classical forgetting-curve models^{[13][14]} and aligns with Tulving's cue-dependent forgetting^[15].

Structural importance:

$$C_{\text{structural}} = 0.4 \cdot \text{centrality} + 0.4 \cdot \text{user_priority} + 0.2 \quad (4)$$

Recency boost:

$$C_{\text{recency}} = 1 + 0.5 \cdot e^{-0.3 \cdot \Delta t} \quad (5)$$

A node is archived once $C_t < 0.25$ for more than 50 consecutive turns.

3.5 The Cue Engine

Before generation, the system scores active nodes by $\text{similarity} \times \text{confidence}$, with a $1.2\times$ priming multiplier. If the best score exceeds an adaptive threshold τ and a minimum turn-gap has elapsed, the engine fires a cue.

Table 2: The Six Cue Types

Cue Type	Trigger	Purpose
Recognition	High similarity, no conflict	Gentle recall, low pressure
Contrast	New input contradicts existing node	Surfaces conflict for human resolution
Correction	Low confidence, high similarity	System admits fallibility
Continuation	Session re-entry, high centrality	Re-onboarding after a gap
Branching	Moderate similarity near existing node	Prevents conflation
Elaboration	Vague human reference	Grounds ambiguous mentions

3.6 Progressive Hydration

On a successful cue, the system starts at the cheapest retrieval level and escalates only on explicit request: **Summary** (~ 50 tokens, no vector access), **Expansion** (~ 400 tokens), or **Full** ($\sim 2,000$ tokens). This is the mechanism responsible for MSG's token economy.

3.7 Socratic Garbage Collection

When an update contradicts a node with more than two dependents, MSG pauses and surfaces the conflict via a Contrast cue. The update only executes after human confirmation. This shifts conflict resolution onto the party with the most context.

Key Insight

Socratic GC is not merely a safety mechanism — it is a *structural* feature. By routing contradictions of high-centrality nodes through human confirmation, MSG ensures the dependency DAG's validity is maintained by the agent with the most context.

3.8 Safety Layer

Deprecated nodes are never deleted (tombstoned). All memory updates are logged for audit. Deprecating high-centrality nodes requires explicit confirmation (poisoning defense). The system is architecturally barred from silently overwriting established facts (gaslighting prevention).

4. Experimental Methodology

Design Rationale. We isolated three structural mechanisms testable without a live LLM or human subjects: update-in-place deduplication, cue-gated token economy, and DAG cascade invalidation.

World Generation. Each world instance randomly draws: topic count (15–40), conversation length (250–600 turns), embedding dimensionality (24–64), DAG edge density (0.25–0.55). Topics are assigned to five heterogeneous categories with distinct priority ranges and contradiction multipliers.

Turn Generation. Each turn is create/update/distractor/query with randomized mixture. Token costs follow a three-component mixture (short chat 65%, explanation 25%, heavy paste 10%).

Baselines. *Full context* (oracle, accuracy = 1.0, linear cost). *Naive RAG* (indexes every chunk without versioning).

MSG-lite. Implements the confidence model exactly as specified, including PageRank centrality recomputed every 50 turns. All six cue types with differentiated acceptance probabilities.

Limitation

The cue-type acceptance rates are hand-set, not measured. This is a central limitation and the target of the LLM-in-the-loop pilot.

Reproducibility. 11 independent runs, each with 12 base instances + 6 sweep points (0.05–0.40 contradiction rate). All parameters resampled per instance. Master seeds reported for exact reproduction.

5. Results

5.1 Base Condition (11 runs × 12 instances)

Table 3: Base Condition Results (132 randomized world instances)

System	Accuracy (mean ± std)	Tokens / turn
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Full context	1.000 ± 0.000	$54,736 \pm 4,798$
Naive RAG	0.303 ± 0.030	195 ± 14
MSG	0.531 ± 0.050	41 ± 5

MSG's accuracy advantage: **+22.8 percentage points** (range +13.4 to +27.7 pp). Token efficiency: **4.8 $\times$ fewer tokens** than RAG. Token consumption remained approximately **three orders of magnitude** lower than full context.

5.2 Contradiction-Rate Sweep

Table 4: Contradiction-Rate Sweep (11 runs aggregated)

Churn Rate	RAG Acc.	MSG Acc.	Gap	Eff. Ratio
0.05	0.428	0.632	+0.204	4.13 $\times$
0.10	0.397	0.590	+0.193	4.28 $\times$
0.15	0.377	0.575	+0.198	4.68 $\times$
0.20	0.315	0.517	+0.202	5.09 $\times$
0.30	0.232	0.452	+0.220	5.45 $\times$
0.40	0.112	0.413	+0.301	5.63 $\times$

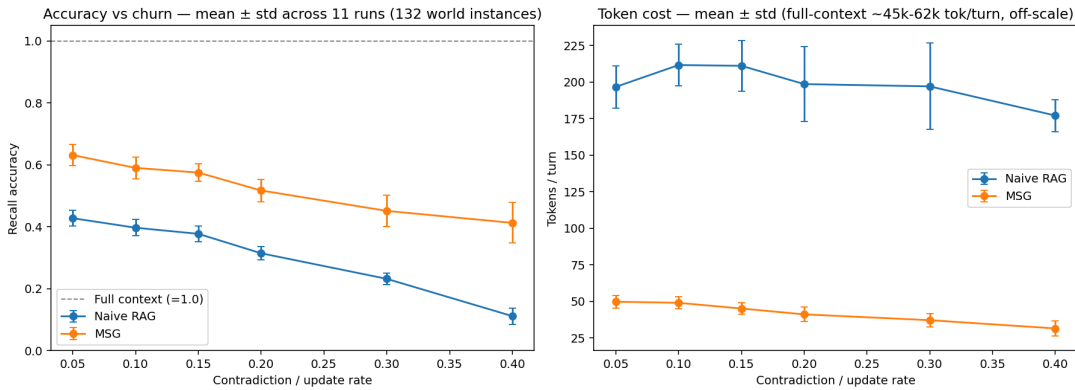


Figure 1: Left: MSG's accuracy advantage over naive RAG widens as contradiction rate increases. Right: Token costs with MSG remaining substantially cheaper. Full context ($\sim 45k$ – $62k$ tokens/turn) omitted as off-scale. Error bars show ± 1 std across 11 runs.

Consistency check: Across all 6 sweep points $\times$ 11 runs (66 comparisons), MSG exceeded RAG in accuracy in **every single case** — zero exceptions. The gap widens with churn, the signature prediction of update-in-place.

5.3 Cascade Invalidation Check

In a minimal two-node test ($A \rightarrow B$), updating A discounted B 's confidence from 0.700 to 0.630 (Equation 1). Naive RAG has no analogous signal: a chunk representing B is retrieved with identical similarity regardless of whether A has changed.

5.4 LLM-in-the-Loop Pilot

Table 5: LLM-in-the-Loop Pilot Results (Claude Sonnet judge)

#	Cue Type	Control	Verdict
1	Recognition	Current	HIT
2	Recognition	Current	HIT
3	Recognition	Stale	CORRECTION
4	Contrast	Stale	CORRECTION
5	Branching	Current	HIT
6	Recognition	Current	HIT

Result: 6/6 correct discriminations — all four current cues accepted, both stale negative controls correctly flagged. A naive “approve anything plausible” judge would have scored 4/6 at best.

6. Discussion

RAG's vulnerability is *structural*, not incidental. Because RAG indexes every version without marking supersession, its accuracy decreases with churn rate ($0.428 \rightarrow 0.112$, a 74% relative decline). MSG's update-in-place removes the possibility of retrieving a stale cued node, so its decline is far shallower ($0.632 \rightarrow 0.413$, a 35% relative decline). The widening gap is the signature prediction of the update-in-place hypothesis.

6.1 Cognitive Foundations

MSG's cue engine maps onto five established findings in cognitive psychology: (1) **Encoding specificity**^[16] — cues authored at creation time match their encoding context. (2) **Spreading activation**^{[10][11]} — the association graph operationalizes priming. (3) **Recognition vs. recall**^{[17][18]} — MSG poses recognition tasks,

not recall demands. (4) **Retrieval-induced forgetting**^[19] — flagged as a testable risk. (5) **Metacognition**^[20] — confidence scores as machine-side feeling-of-knowing estimates.

6.2 Scrutiny

Two results deserve scrutiny: (1) The false-cue rate is stable at $\sim 24\%$ across all conditions, suggesting it is driven more by fixed acceptance probabilities than the confidence mechanism. (2) Socratic-GC events scale sharply with contradiction rate (up to 12.2 in individual runs), implying MSG's cost advantage has a ceiling under very high churn.

7. Implications

We offer these as candidate directions worth testing, not claims. **Agent memory:** cue-gated retrieval suggests contradiction resolution as a design axis. **Coding copilots:** dependency-rich sessions could benefit from structural conflict surfacing. **Personal assistants:** confirmation before overwriting preferences is plausible but introduces interaction-design costs. **Local LLMs:** token economy matters most where compute is constrained. **Enterprise systems:** tombstone design doubles as compliance audit trails.

8. Limitations and Threats to Validity

1. **Synthetic embeddings.** All embeddings are Gaussian-noise perturbations, not real sentence embeddings. Retrieval behavior could differ under real embeddings.
2. **Rule-based acceptance.** The pilot is $n = 6$ on a scenario written by the architecture's designers — a real bias source.
3. **No comparison against real baselines.** We compare against naive RAG, not MemGPT/Letta, Mem0, Zep, or Mnemosyne as implemented.
4. **No standard benchmark.** LongMemEval and LoCoMo were not used; all results are on a synthetic distribution.
5. **No per-mechanism ablation.** All mechanisms run simultaneously; marginal contributions are not isolated.
6. **Hand-tuned thresholds.** $\tau_0 = 0.30$ is chosen by inspection, not learned.
7. **Acceptance rates are assumptions.** Cue-type acceptance rates are hand-set, not measured.
8. **Unmeasured contradiction rates.** The simulated range (0.05–0.40) is for sweep coverage, not calibrated to real conversations.
9. **Retrieval-induced forgetting unmodeled.** Identified but not measured.

10. **No multi-turn dialogue evaluation.** Measures per-query accuracy, not conversational coherence.
11. **No latency measurements.** Token cost does not capture wall-clock performance.

9. Future Work

1. **Replace synthetic embeddings** with real sentence embeddings and re-run the full sweep.
2. **Scale the LLM-in-the-loop pilot** to independently-authored scenarios across multiple models.
3. **Build a working implementation** and evaluate on LongMemEval and LoCoMo against real baselines.
4. **Run per-mechanism ablations** to isolate marginal contributions.
5. **Learn adaptive thresholds** via bandit-style policies.
6. **Conduct a human-subjects pilot** to validate cue-type acceptance assumptions.

10. Conclusion

We introduced the Mnemonic State Graph, a memory architecture premised on treating the human as an active, cue-triggerable memory store. In a controlled synthetic evaluation spanning 132 randomized world instances across 11 independent runs, the core structural mechanisms produced a consistent, non-shrinking accuracy advantage over naive RAG (mean +22.8 percentage points, holding in 66/66 comparisons) at roughly one-fifth the token cost. A companion pilot found correct discrimination on all six cue judgments (6/6).

These results validate the mechanism, not the system. We view this work as the necessary first step — a stress-tested design and a reproducible simulation harness — before committing the engineering effort to a full implementation. The broader question MSG raises — whether distributed cognition between human and machine can be systematically leveraged as a design principle — remains open. This paper provides one architecture and one encouraging set of synthetic results. The next step is to find out whether they hold in the real world.

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Appendix: Reproducibility

The full simulation harness (`msg_simulation.py`) and raw per-run JSON output for all 11 runs accompany this paper and are published in the GitHub repository. Every run reports its master seed; `python3 msg_simulation.py --seed <value>` reproduces any specific run exactly.

A.1 GitHub Repository

All code, data, figures, and license terms are published at:

<https://github.com/agnel/msg-memory-architecture>

- **Simulation:** `msg_simulation.py` — seed-reproducible harness.
- **Datasets:** Raw per-run JSON for all 11 runs and aggregated summary.
- **Figures:** Plotting scripts generated from aggregate JSON.
- **License:** See `LICENSE` in repository root.

A.2 Simulation Parameters

Table A1: Fixed Simulation Parameters

Parameter	Value	Description
<code>SUMMARY_TOKENS</code>	50	Base cost for summary-tier cue
<code>EXPANSION_TOKENS</code>	400	Cost for expansion-tier retrieval
<code>FULL_TOKENS</code>	2,000	Cost for full-tier retrieval
<code>LAMBDA</code>	0.95	Base confidence decay per missed cue
<code>REGENCY_HALFLIFE</code>	10	Recency boost half-life (turns)
<code>SOCRATIC_ASK_COST</code>	35	Token cost of Socratic-GC question
<code>RAG_TOPK</code>	3	Chunks retrieved by naive RAG
<code>base_tau</code>	0.30	Base adaptive cueing threshold
<code>min_gap</code>	4	Minimum turns between cues

A.3 Data Availability

All data reported are derived from `msg_aggregate_summary_11runs.json`. Raw per-run JSON files total approximately 2.3 MB. The simulation harness, aggregate data, and pilot artifact are sufficient to reproduce every number, table, and figure.

A.4 Computational Requirements

Python 3.12 with NumPy 2.0, NetworkX 3.3, and Matplotlib 3.9. A single run completes in 30–45 seconds on a standard laptop. The 11-run replication completes in under 10 minutes. Memory requirements are <500 MB.